

HFSHD in JOREK model600: what drives the divertor potential, and what is still missing

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Abstract

Two independent numerical experiments show that the high-field-side density asymmetry in JOREK `model600` is controlled by the product $\eta j_{\parallel}$ in the SOL and private flux region, exactly as the SOLPS mechanism of Senichenkov *et al.* (CPP **62**, 2022, e202100177) requires. Raising $\eta j_{\parallel}$ by either route — restoring a volumetric current term in the SOL, or raising the resistivity directly — produces the divertor potential structure and a high-field-side density enhancement; removing it removes both. This reverses the campaign’s earlier conclusion that the mechanism was absent from the model.

What has *not* been reproduced is the full HFSHD signature. Both experiments produce a density enhancement localised *at* the inner target; neither produces the cold, dense column extending from the target up the inner leg that defines HFSHD. The reason is identified: both routes amplify the *electrical* half of the loop and neither supplies the *thermal* half, because artificial resistivity does not cool the plasma. Section 6 states precisely which claims are measured and which are inferred; several confident interpretations made during this work were falsified by the data and are recorded in Section 7 so they are not revisited.

1 The mechanism being tested

Senichenkov’s chain, in the order the causality runs:

$$\partial_{Zp}|_{\text{PFR}} \longrightarrow \nabla \cdot \mathbf{j}_{\text{dia}} \longrightarrow \nabla_{\parallel} j_{\parallel} \longrightarrow E_{\parallel} = \eta j_{\parallel} \longrightarrow \Phi \text{ structure} \longrightarrow \mathbf{v}_E \longrightarrow n_{\text{HFS}} \quad (1)$$

The divergent diamagnetic current in the private flux region must close as parallel current to the plates; driving that current through cold, resistive plasma costs a parallel electric field; the integrated field is a potential structure near the X-point; its $\mathbf{E} \times \mathbf{B}$ carries plasma from the outer to the inner divertor. The loop then closes on itself, because the transported density cools the inner leg, raising η and therefore $E_{\parallel}$.

1.1 Every ingredient is present in model600

The vorticity equation *is* $\nabla \cdot \mathbf{j} = 0$, so the second and third arrows of (1) are satisfied by construction rather than approximately:

SOLPS ingredient	model600 term	source line
$\nabla \cdot \mathbf{j}_{\text{dia}}$	<code>u_Eq_grad_p</code> , $R^2(v_{sp0t} - v_{tp0s}) \rightarrow -2R \partial_{Zp}$	<code>mod_elt_matrix_fft:1593</code>
$\nabla_{\parallel} j_{\parallel}$	<code>u_Eq_JxB</code> , $[\psi, zj]$	<code>:1586</code>
$-\nabla_{\parallel} p_e / (en_e)$	$\tau_{\text{IC}} \cdot 2 / (r_0 B^2)$ bracket	<code>:1563</code>
$-\alpha_e \nabla_{\parallel} T_e / e$	thermoelectric term	<code>:1571</code>
$j_{\parallel}$ through the plate	weak Galerkin j - V sheath BC	<code>mod_sheath_trace</code>

and the switches are live in the production input: `tauc = 5.388e-3`, `eta_T_dependent = .true.` (default), `keep_current_confine_strength = 1` (default). No term of the mechanism is switched off.

2 Result 1: the cutoff scan

`keep_current_prof` maintains the current profile through a source zj_{src} . The parameter `keep_current_confine_strength` masks that source outside the confined region: 0 leaves it acting everywhere including the SOL and PFR, 1 removes it there entirely. Four runs, identical but for this parameter, compared at the same cycle and time ($1000, t = 2.83 \times 10^{-3} \text{ s}$):

strength	0	0.25	0.5	1
u_{max} (read from plot)	39.5	25.4	20.7	12.4
PFR current	broad, filling	weaker	weaker	near zero
inner-leg density lobe	large, $n_{e20} \approx 1$	smaller	elongated	absent

Three quantities, monotonic, in the same direction, across four points. The potential amplitude changes by only $3.2\times$ while the density lobe goes from prominent to absent — the density response is strongly nonlinear, which is the signature of a threshold rather than a proportional response.

2.1 Why this is causation and not correlation

`keep_current_prof` enters *only* the ψ equation, as $\eta(zj - zj_{\text{src}})$. It appears nowhere in the continuity equation. Its only path to the density is therefore

$$zj_{\text{src}} \rightarrow \text{EMF in Ohm's law} \rightarrow \Phi \rightarrow v_E \rightarrow n. \quad (2)$$

That is a statement about which terms exist in which equations, not an inference from four correlated panels. (A second path exists in principle — the source changes $zj = \Delta^*\psi$, hence the field, hence parallel transport — but SOL currents are $\sim 0.1\%$ of I_p , so it is negligible.)

2.2 Why a boundary condition cannot substitute

This is the structural reason the converged sheath BC, on its own, did not produce HFSHD.

A boundary condition drives current that flows *through* the domain: in at one plate, out at the other. Along a field line $j_{\parallel}$ is then roughly constant, so $\int \eta j_{\parallel} dl$ is **monotonic** — a target-to-target bias.

A potential hill is a *local maximum*: Φ falls from the X-point toward *both* plates, so $j_{\parallel}$ must flow *out of* the X-point region in both directions and therefore **change sign** along the field line. A current that changes sign requires a source in between. No boundary condition can supply that; only a volumetric $\nabla \cdot \mathbf{j}$ term can.

This is measurable in the runs, not merely argued. With the sheath BC active the u field shows $\Phi > 0$ on the inner leg and $\Phi < 0$ on the outer — a monotonic target-to-target bias with no local extremum at the X-point. And an imposed bias of that shape was separately measured to be a weak lever: **36 V moved the integrated in-out density ratio from 1.171 to 1.189**, i.e. 5×10^{-4} per volt.

3 Result 2: resistivity alone, with no artificial source

The obvious objection to Section 2 is that zj_{src} is not a physical current. The second experiment removes it. With `keep_current_confine_strength` = 1 (source fully masked in the SOL), the sheath BC and thermoelectric term active, and η raised $100\times$:

	$\eta \times 1$	$\eta \times 100$
$u_{\max}$	12.4	108.4
PFR potential structure	thin wedge	strong, localised, closed
current in the divertor	thin filaments on the separatrix	broad, diffuse
HFS density enhancement	absent	present at the target

The structure appears *without any artificial current in the SOL*. Whatever current the model generates physically, multiplied by a large enough η , produces the potential structure and an HFS density enhancement. This is the same chain as Section 2 reached by an independent route, and it is the stronger of the two results because nothing is imposed.

Two subsidiary observations:

- The current is no longer filamentary. At $\eta \times 1$ it forms thin layers hugging the separatrix; at $\eta \times 100$ resistive diffusion is $100\times$ faster and those layers broaden past the grid scale. The filaments are consistent with flux dragging against a frozen ψ (`sheath_wall_vel = 0`), which suggests that parameter as a way to broaden them *without* unphysical resistivity.
- Scaling: $u_{\max}$ rose by $8.7\times$ for $100\times$ in η — sublinear, as expected since $j_{\parallel}$ responds. Working backwards, reaching the $u_{\max} \approx 40$ that produced a clear lobe in Section 2 requires roughly $\eta \times 10\text{--}20$.

That factor is available physically. Since $\eta \propto T_e^{-3/2}$, cooling the leg from the present $\approx 25\text{ eV}$ to $\approx 5\text{ eV}$ gives $\eta \times 11$. The required amplification does not need artificial resistivity; it needs a moderately detached leg.

4 What is still missing: the thermal half

Neither experiment reproduced HFSHD. Both produced a density enhancement *at* the inner target. HFSHD is more than that: it is a cold, dense structure extending *from* the HFS target *up* the inner leg, past the X-point, into the HFS SOL. In the runs, the inner leg immediately above the enhancement is among the most dilute regions in the domain.

$\eta \times 100$ is not equivalent to a cold leg. It reproduces the resistivity of a $\approx 1.2\text{ eV}$ plasma *for Ohm's law only*. Every temperature-dependent process — line radiation, ionisation balance, volume recombination, charge-exchange momentum loss, parallel conduction — still sees 25 eV . The upward extension of HFSHD is a detachment front, and fronts propagate on radiation and power balance, not on $\mathbf{E} \times \mathbf{B}$. A run with artificial η therefore *cannot* produce the column, however large the potential structure becomes.

This separates the loop cleanly:

half	supplies	status
electrical	$\eta j_{\parallel} \rightarrow \Phi \rightarrow \mathbf{v}_E \rightarrow$ seeds the asymmetry	demonstrated, twice
thermal	radiation, recombination $\rightarrow$ cold column up the leg	not present

The thermal half requires the leg to be genuinely cold, which requires recycling and radiation — and in ASDEX Upgrade HFSHD is observed in *pure deuterium* high-density discharges, so hydrogenic radiation, CX momentum loss and volume recombination are sufficient. Impurity seeding is not a prerequisite.

The high-density deuterium runs already produce dense legs, but *symmetric* ones. The electrical seed and the thermal amplifier have so far never been present in the same run: the

low-puff cases have the potential structure available (via artificial η) but warm legs; the high-puff cases have cold legs but no measured potential structure.

5 Numerical suppression: a specific, untested suspect

One direct comparison in the run set shows a density lobe present and then absent:

configuration	inner-leg lobe
cutoff	present, $n_{e20} \approx 0.5$
cutoff + sheath j + <code>D_perp_sc_num</code> = 10	absent

Two things changed at once; this comparison cannot attribute the loss. Either the sheath BC's $\mathbf{E} \times \mathbf{B}$ transports the lobe away, or the shock-capturing diffusion erases it.

The second is the more targeted hypothesis. `D_perp_sc_num` defaults to 0 and is set to 10; shock-capturing diffusion acts precisely where gradients are steep, and HFSHD *is* a steep gradient — dense HFS against dilute LFS across a sharp boundary through the PFR. It was introduced to suppress cell-Péclet ringing in earlier floating-potential work; whether it also removes the structure of interest is untested and is resolved by one run with `use_sc` = `.false`.

6 Epistemic status

Several confident interpretations made during this work were falsified within hours. The following distinguishes what the data support from what is inferred.

claim	status
Density asymmetry, PFR current and $u_{\max}$ all scale monotonically with the SOL current source	Measured , 4 points, matched time
The source reaches the density only through Φ	Structural : it appears only in the ψ equation
A boundary condition produces a monotonic bias, not a local maximum	Structural , and consistent with the measured u topology
An imposed target-to-target bias is a weak lever	Measured : 5×10^{-4} per volt
At $\eta \times 100$ with no artificial source, the potential structure and an HFS enhancement appear	Measured , but the run <i>crashed at 610 steps</i> and is not converged ($t = 2.2 \times 10^{-4}$ s)
$\eta \times 10$ –20 suffices, and a 5 eV leg supplies it	Inferred from a sublinear two-point scaling; not verified
zj_{src} is a proxy for the physical plate-closing current	Structural analogy only . zj_{src} is the initial equilibrium profile, not the divergent $\nabla \cdot \mathbf{j}_{\text{dia}}$ pattern. The scan shows a volumetric current term drives the chain; it does <i>not</i> establish that the physical current does so at its physical magnitude
The thin current layers are flux dragging against frozen ψ	Hypothesis , supported by their disappearance at high η ; <code>sheath_wall_vel</code> untested
D_perp_sc_num erases the lobe	Confounded ; two changes at once
The inner-leg lobe is drift-fed rather than puff-fed	Not established . <code>valves(1)</code> sits at R 1.3–1.4, Z – 1.11–1.05, directly beneath it
Sign convention of Φ relative to the plotted u	Unresolved . $\Phi = -F_0 u$, so red u is negative Φ ; <code>postproc</code> ’s <code>Phi</code> expression additionally returns $+F_0 u$. Pin this before comparing topology to SOLPS

Supporting measurements from the same campaign

- Φ is **not** a flux function: flux-weighted misalignment 7.2° , with closed u contours (local extrema) in the PFR where ψ has none.
- Cross-field transport in the PFR: $\langle |\Gamma_\psi| \rangle / \langle |\Gamma_{\parallel, \text{pol}}| \rangle = 0.168$ with net fraction 0.558, i.e. $\approx 9\%$ of the parallel flux crossing flux surfaces and *not* cancelling.
- The reduced-MHD balance $v_\psi \approx \eta j_\parallel / |\nabla \psi|$ is verified: 21 m/s measured against 15 m/s predicted. The model is in the correct resistive balance.
- The thermoelectric term contributes +2% to the inner sheath potential, *reduces* the in–out asymmetry by 45%, and its changes avoid the PFR (concentration ratio 0.33). It is not the driver.

7 Falsified during this work

Recorded so they are not revisited:

- “*The divertor is too cold; Φ is $40\times$ too small.*” The low-puff legs are 20–30 eV. The 1 eV figure came from a different case with a broken pump.
- “*The resistivity floor is clamping η .*” `t_min_neg = 3e-5` puts the knee at 1.93 eV, far below the leg temperature. Inactive.
- “ *F_0 has the wrong sign.*” The wiki is explicit that standard ASDEX Upgrade is $F_0 > 0$ in JOEREK; the toroidal angle runs opposite to the device convention. The remaining condition `Psi_axis < Psi_xpoint` has not been checked.
- “*The thermoelectric term is the key.*” 2%, wrong direction, PFR-avoiding.
- “*The sheath BC alone can build the hill.*” Structurally it cannot; see §2.3.
- “*Impurity seeding is required.*” HFSDH is observed in pure-D high-density ASDEX Upgrade discharges.
- “*The uncut keep_current_prof was 100% artifact.*” What was falsified was the η -scaling interpretation. The source acts as a volumetric EMF in the SOL and drives the chain; see the caveat in §6 on how far that analogy extends.

8 Next steps

Ordered by information per unit compute.

1. **High-D with `use_sc = .false.`** One line. The high-density case already supplies the thermal half; this tests whether the structure is being erased as it forms. Step $10 \rightarrow 3 \rightarrow 1$ if ringing returns. *This is the only run in which both halves of the loop could be present at once.*
2. **High-D with `eta = 2.e-7` ($\times 10$).** Insurance: guarantees the electrical half is above threshold while the cold leg supplies the thermal half. Neither half faked.
3. **`sheath_wall_vel = 4.e-3`.** Tests whether the thin current layers are flux dragging. If they broaden and current enters the PFR interior, the amplification is recovered without unphysical η . Mesh-dependent — scan it.
4. **`valves(1) off`.** Decides whether the inner-leg lobe is drift-fed or puff-fed. Until this is answered it is not established that the lobe is HFSDH at all.
5. **Fix the polygons** before any detachment attempt: all four pump and valve quads are specified in bowtie order (bottom-left, top-left, bottom-right, top-right), which self-intersects and roughly halves the effective areas. Swap the last two vertices of each, or set `only_in_polygon = .false.` and use a whole-wall albedo.

What would close the question

The measurement that distinguishes “mechanism present but subcritical” from “mechanism absent” is $\Delta\Phi$ along the field line from the X-point to each strike point, compared against $\int \eta j_{\parallel} dl$ over the same path. SOLPS obtains ~ 100 V. Tooling exists (`util/postproc_hill.in`, `util/analyse_hill.py`) and has not been run.

Runs: AUG #38773, `model600`, boundary type 1, $n_{\text{tor}} = 1$, branch `thermoelectric-ohm`. Reference: I. Senichenkov *et al.*, Contrib. Plasma Phys. **62** (5–6), e202100177 (2022). Companion documents: `doc/sheath_bc_whitepaper.pdf` (the weak Galerkin j -V sheath BC), `doc/jorek_vpar_bc_drifts.pdf` (the Mach-1 BC and its drift term).